Regression through the origin

A no-intercept regression model often seems appropriate in analyzing data from chemical and other manufacturing processes.

The no-intercept model is

$$y = \beta_1 x + \epsilon$$

Given n observations $(y_i, x_i), i = 1, 2, ..., n$ the least squares function is

$$S(\beta_1) = \sum_{i=1}^{n}(y_i - \beta_1 x_i)^2$$

The only normal equation is

$$\hat{\beta}_1 \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} y_i x_i$$

and the least squares estimator of the slope is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i}{\sum_{i=1}^{n} x_i^2}$$

The estimator of $\hat{\beta}_1$ is unbiased for β_1 and the fitted regression model is

$$\hat{y} = \hat{\beta}_1 x$$

The estimator of σ^2 is

$$\hat{\sigma^2} = MS_{Res} = \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n-1} = \frac{\sum_{i=1}^{n} y_i^2 - \hat{\beta}_1 \sum_{i=1}^{n} y_i x_i}{n-1}$$

The $100(1-\alpha)\%$ on β_1 is

$$\beta_1 \pm t_{\frac{\alpha}{2}, n-1} \sqrt{\frac{MS_{Res}}{\sum_{i=1}^{n} x_i^2}}$$

Example in R

```
# reading in the data
Y <- c(2158.70, 1678.15,2316,2061.30,2207.50,1708.30, 1784.70,1784.70,2575,2357.90)

X <- c(15.50,23.75,8.00,17,5.5,19,24,2.5,7.5,11)

# fitting the model
```

```r
model2 <- lm(Y~-1+X)

summary(model2)
```

Estimation by Maximum Likelihood

Consider the data $(y_i, x_i), i = 1, 2, 3, ..., n$. If we assume that the errors in the regression model are $NID(0, \sigma^2)$ then the observations y_i in this sample are normally and independently distributed random variables with mean $\beta_0 + \beta_1 x$ and variance σ^2.

The likelihood function is found from the joint distribution of the observations.

For the simple linear regression model with normal errors the **likehood function** is

$$\begin{aligned} L(y_i, x_i, \beta_0, \beta_1, \sigma^2) &= \prod_{i=1}^{n} (2\pi\sigma^2)^{-1/2} exp\Big[-\frac{1}{2\sigma^2}(y_i - \beta_0 - \beta_1 x_i)^2 \Big] \\ &= (2\pi\sigma^2)^{-\frac{1}{2}} exp\Big[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2 \Big] \end{aligned} \tag{11}$$

The maximum likelihood estimators are the parameter values say $\tilde{\beta_0}, \tilde{\beta_1}$ and $\tilde{\sigma^2}$ that maximize L or equivalently.

In L thus:

$$\begin{aligned} \ln L(y_i, x_i, \beta_0, \beta_1, \sigma^2) &= -\Big(\frac{n}{2}\Big) \ln 2\pi - \Big(\frac{n}{2}\Big) \ln \sigma^2 \\ &= -\Big(\frac{1}{2\sigma^2}\Big)\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2 \end{aligned} \tag{12}$$

and the maximum likelihood estimators $\tilde{\beta_0}, \tilde{\beta_1}$ and $\tilde{\sigma^2}$ must satisfy

$$\frac{\partial \ln L}{\partial \beta_0}|_{\tilde{\beta_0}, \tilde{\beta_1}, \tilde{\sigma^2}} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(y_i - \tilde{\beta_0} - \tilde{\beta_1} x_i) = 0$$

$$\frac{\partial \ln L}{\partial \beta_1}|_{\tilde{\beta_0}, \tilde{\beta_1}, \tilde{\sigma^2}} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(y_i - \tilde{\beta_0} - \tilde{\beta_1} x_i)x_i = 0$$

$$\frac{\partial \ln L}{\partial \sigma^2}|_{\tilde{\beta_0}, \tilde{\beta_1}, \tilde{\sigma^2}} = -\frac{n}{\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(y_i - \tilde{\beta_0} - \tilde{\beta_1})^2 = 0$$

The solution to the above equations gives the **maximum likelihood estimators**

$$\tilde{\beta_0} = \bar{y} - \tilde{\beta_1}\bar{x}$$

$$\tilde{\beta}_1 = \frac{\sum_{i=1}^{n} y_i(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

$$\tilde{\sigma^2} = \frac{\sum_{i=1}^{n} (y_i - \tilde{\beta}_0 - \tilde{\beta}_1 x_i)^2}{n}$$

In general maximum-likelihood estimators have better **statistical properties** than least-squares estimators.

The maximum-likelihood estimators are unbiased and have minimum variance when compared to all other unbiased estimators.

They are also consistent estimators.

Example 3

The purity of oxygen produced by a fractional distillation process is thought to be related to the percentage of hydrocarbons in the main condensor of the processing unit. Twenty samples are shown below.

Purity (%)	**Hydrocarbon (%)**
86.91	1.02
89.85	1.11
90.28	1.43
86.34	1.11
92.58	1.01
87.33	0.95
86.29	1.11
91.86	0.87
95.61	1.43
89.86	1.02

Table 2: Purity and Hydrocarbon percentages

i Fit a simple linear regression model to the data

ii Test the hypothesis $H_0 : \beta_1 = 0$

iii Calculate R^2

iv Find a 95% CI on the slope

v Find a 95% CI on the mean purity when the hydrocarbon percentage is 1.00